



Set Problem Bank — HashSet, LinkedHashSet, TreeSet (Java)

Legend

★ = Core Placement Problem (Mandatory)

Covers internal implementation details alongside application-level problems.

These problems should be completed by every student before moving to Map.

Section 1: Set Basics and Creation

- ★ Create and Display a HashSet
- ★ Create and Display a LinkedHashSet
- ★ Create and Display a TreeSet
- ★ Create a Set from a List (Remove Duplicates)

Find Size of a Set using size()

Create an Immutable Set using Set.of()

Section 2: HashSet vs LinkedHashSet vs TreeSet Internals (Core Concept)

- ★ Internal Structure of HashSet (Backed by HashMap, Hashing)
- ★ Internal Structure of LinkedHashSet (Hash Table + Linked List for Order)
- ★ Internal Structure of TreeSet (Red-Black Tree, Sorted Order)
- ★ Demonstrate Insertion Order Loss in HashSet
- ★ Demonstrate Insertion Order Preservation in LinkedHashSet
- ★ Demonstrate Natural Ordering in TreeSet
- ★ Time Complexity Comparison: add()/contains() — HashSet vs TreeSet
- ★ Custom hashCode() and equals() Contract for Objects in a HashSet

Custom Comparator for TreeSet Ordering

Section 3: Adding and Removing Elements

- ★ Add a Single Element to a Set
- ★ Add Multiple Elements using addAll()
- ★ Remove an Element using remove()

Remove Elements Matching a Condition using removeIf()

Clear All Elements from a Set

Section 4: Set Operations

- ★ Union of Two Sets
- ★ Intersection of Two Sets
- ★ Difference of Two Sets



Symmetric Difference of Two Sets

★ **Check if a Set is a Subset**

Check if Two Sets are Disjoint using `Collections.disjoint()`

Section 5: Searching and Iterating

★ **Check Membership using `contains()`**

★ **Iterate through a Set using Enhanced For-Loop**

Iterate using Iterator

★ **Navigate a TreeSet using `first()`, `last()`, `higher()`, `lower()`**

Get `headSet()`/`tailSet()`/`subSet()` from a TreeSet

Section 6: Set Conversion and Utility

★ **Convert Set to List and Sort It**

Convert Set to Array

Convert String to a Set of Characters

Create an Unmodifiable Set using `Collections.unmodifiableSet()`

Section 7: Set-Based Logical Problems

★ **Find Unique Elements from Multiple Lists**

★ **Find Common Elements among Three Sets**

★ **Check if Two Strings are Anagrams using Set**

★ **Remove Duplicates while Preserving Order using LinkedHashSet**

Count Unique Words in a Sentence using Set

Section 8: Real World Set Applications

★ **Unique Visitor Tracker using Set**

★ **Common Friends Finder (Social Network Problem)**

Duplicate Email Detector using Set

★ **Tag / Keyword Deduplication System**

★ **Leaderboard with Sorted Unique Scores using TreeSet**